Alternative semantics

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June 24, 2024

1 Singleton-Semantics

1.1 •

$$((\varepsilon x : x \in Q) \rightsquigarrow t) \longleftrightarrow (t \in Q)$$

$$((\varepsilon y : \psi) \rightsquigarrow t) \longleftrightarrow \psi[t/y]$$

2 Appendix 2: Alternative subset semantics [unfinished]

2.1 Logic

$\mathcal{A}$ Alphabet

2.1.1 Syntax

Basic notion of the system is "class".

2.1.2 Terms

Terms ::=

2.1.3 Formulas

Definition 2.1. Fm ::=

Definition 2.2.

$$M(Z) \longleftrightarrow \exists C : Z \in C$$

Gödel notation:

2.1.4 Axioms

interpretation is denoted by I here.
Usually assignment is an univocal correspondence between variables and sets. In our case assignment is an univocal correspondence between variables and conglomerate-singletons of classes.

There are several possible ways of interpreting "in" relation: Assume X and Y are some terms(sic! not just variables). 1) Weak $X \in Y$ is valid iff for any assignment any class-element of I(X) is in any class-element of I(Y).

Hypothesis:

$$A(\varepsilon x : A) \longleftrightarrow \exists x : A(x)$$

Axiom schema
Following class existence metatheorem for NBG, we can introduce an axiom schema.
To begin with let's introduce recursive function on formulas: it substitutes class variables with set variables.
The resulting formula contain only set variables.
If φ contains only set variables, then $\{x|\varphi\}$ is defined.

$$(\varphi \in \text{setify[Fm]}) \overset{meta}{\Longrightarrow} (\vdash (\{x|\varphi\}\downarrow))$$

2.2 Semantics

"super-large" model.
K - conglomerate of all classes.

Assignment:

$$A : \text{Var} \to \mathcal{P}(K)$$

$$I_A^{\text{Fm}}(S\downarrow) \longleftrightarrow (I_A^{\text{Term}}(S) \neq \emptyset)$$

Definiton of truth:

$$\text{Bool} := \{\text{False}, \text{True}\}$$

$$M, A \vDash \varphi \overset{meta}{\Longleftrightarrow} I_A(\varphi) = \text{True}$$

$$(M \vDash \varphi) \overset{meta}{\Longleftrightarrow} (\forall A \in \text{Assignment} : M, A \vDash \varphi)$$